The CTI Universal Law

A First-Principles Study Guide

Everything you need to understand the math,
the evidence, and the open directions.

There is a single equation that tells you how well **any** classifier (neural network, biological brain, anything) will perform, using only the **geometry** of how it organizes information — and it's derivable from first principles.

$$\mathrm{logit}(q_{\mathrm{norm}}) = \alpha\, \kappa_{\mathrm{nearest}} - \beta \log(K-1) + C$$

Devansh · March 2026

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The Setup: What Are We Measuring?

The fundamental question

Given a trained neural network (or a biological brain), how good are its internal representations at distinguishing between categories?

We don't care about the final classifier head. We care about the **geometry of the embeddings** — the spatial layout of how the model organizes data points internally. If two classes are well-separated in embedding space, the model can tell them apart easily. If they're tangled up, it can't.

The probe: 1-Nearest Neighbor (1-NN)

To measure representation quality without adding any learnable parameters, we use the simplest possible classifier: **1-nearest neighbor**.

1-NN Classification

Given a test point $\mathbf{x}$:

1. Find its nearest neighbor in the training set (by Euclidean distance in embedding space)
2. Predict $\mathbf{x}$ has the same label as that neighbor

1-NN accuracy depends **only** on the geometry of the embeddings. No probe capacity. No optimization. Pure geometric signal.

Why not a linear probe? A linear probe mixes geometry with the probe's own optimization. It can mask bad geometry if the probe has enough capacity, or fail with good geometry if the hyperparameters are wrong. 1-NN removes this confounder entirely: the accuracy *is* the geometry.

The Key Quantities

q_{norm}: Chance-Normalized Accuracy

Raw 1-NN accuracy is misleading across different K. With K=2, random guessing gives 50%. With K=100, random gives 1%. We normalize:

Chance-Normalized Accuracy

$$q_{\text{norm}} = \frac{\text{acc}_{\text{1NN}} - 1/K}{1 - 1/K} \tag{1}$$

- $q_{\text{norm}} = 0$: exactly at chance (no geometric signal)
- $q_{\text{norm}} = 1$: perfect classification
- $q_{\text{norm}} < 0$: below chance (rare, adversarial geometry)

Example: If K=10 and 1-NN accuracy = 55%:

$$q_{\text{norm}} = \frac{0.55 - 0.10}{1 - 0.10} = \frac{0.45}{0.90} = 0.50$$

κ_{nearest}: The Geometric Signal-to-Noise Ratio

This is the single most important number in the entire theory.

κ_{nearest} — The Key Quantity

$$\kappa_{\text{nearest}}(k) = \min_{j \neq k} \frac{\|\boldsymbol{\mu}_j - \boldsymbol{\mu}_k\|}{\sigma_W \sqrt{d}} \tag{2}$$

Numerator: distance to the *nearest* other class centroid (the hardest competitor)
Denominator: expected within-class L2 radius (isotropic null)
Result: dimensionless signal-to-noise ratio for class separation

Where:

- $\boldsymbol{\mu}_k$ = centroid (mean embedding) of class k
- $\min_{j \neq k}$ = take the **nearest** other class (hardest competitor)
- σ_W = pooled within-class per-dimension standard deviation
- d = embedding dimension

Why "nearest"?

Because in a 1-NN race, the class you're most likely to confuse a point with is the **nearest other class**. Far-away classes are irrelevant. This is the *bottleneck* — the weakest link.

Intuition

Think of κ_{nearest} as a "contrast ratio":

- $\kappa \ll 1$: noise swamps signal → classification at chance
- $\kappa \sim 1$: signal and noise comparable → transition zone
- $\kappa \gg 1$: signal dominates → near-perfect classification

Concrete example

Pythia-160M on AGNews (K=4), final layer: centroid distances between news categories vary by pair. The nearest pair: $\|\boldsymbol{\mu}_{\text{sports}} - \boldsymbol{\mu}_{\text{world}}\| = 15.2$. With $\sigma_W\sqrt{768} = 14.3$: $\kappa_{\text{nearest}} = 15.2/14.3 = 1.06$.

K: Number of Classes

K enters through $\log(K-1)$ because more classes = more competitors. Doubling classes doesn't double difficulty — it increases it *logarithmically* (from the Gumbel extreme-value structure).

Why Logit? Why Linear?

The logit transformation

The logit maps probability $p \in (0, 1)$ to $\mathbb{R}$:

$$\text{logit}(p) = \log \frac{p}{1-p} \tag{3}$$

$$\sigma(x) = \frac{1}{1+e^{-x}} \quad \text{(inverse: sigmoid)} \tag{4}$$

The Law in Two Forms

Logit form:

$$\text{logit}(q_{\text{norm}}) = \alpha\, \kappa_{\text{nearest}} - \beta \log(K-1) + C$$

Equivalent sigmoid form:

$$q_{\text{norm}} = \frac{1}{1 + \exp\big(-(\alpha\, \kappa_{\text{nearest}} - \beta \log(K-1) + C)\big)}$$

This is NOT a logistic regression. It's a structural prediction from extreme value theory. The sigmoid appears because of the Gumbel race probability structure.

The Gumbel Race: Where the Law Comes From

This is the core derivation. Seven steps from first principles.

Step 1: 1-NN as a race

When classifying query $\mathbf{x}$ from class k using 1-NN:

The 1-NN Race

Define:

$$M_+ = \min_{i:\, y_i = k,\, i \neq x} \|\mathbf{x} - \mathbf{x}_i\| \quad \text{(nearest same-class)} \tag{5}$$

$$M_j = \min_{i:\, y_i = j} \|\mathbf{x} - \mathbf{x}_i\| \quad \text{(nearest from class } j\text{, for } j \neq k) \tag{6}$$

1-NN is correct $\iff M_+ < \min_{j \neq k} M_j$
The correct class must "win the race" by having the smallest minimum distance.

Step 2: Extreme Value Theory (EVT)

The minimum of n i.i.d. random variables, after centering and scaling, converges to one of exactly three universal limit distributions (Fisher–Tippett theorem, 1928):

1. **Gumbel** — exponential-type tails (Gaussian, etc.) ← **our case**
2. **Fréchet** — heavy tails
3. **Weibull** — finite support

For Gaussian/sub-Gaussian class conditionals (which neural embeddings approximately satisfy), minimum distances are in the **Gumbel domain of attraction**.

The Gumbel Distribution

$$P(G \leq x) = \exp\big(-\exp(-(x-\mu)/s)\big)$$

μ = location (decreases as $O(\sqrt{\log n})$ with more samples)
s = scale (stays roughly constant)

Step 3: Setting up the Gumbel race

After EVT normalization:

$$M_+ \sim \text{Gumbel}(\mu_+, s) \quad \text{(same-class nearest)} \tag{7}$$

$$M_j \sim \text{Gumbel}(\mu_j^{(-)}, s) \quad \text{(class-}j\text{ nearest)} \tag{8}$$

Key: all Gumbels share approximately the **same scale** s (within-class spread is similar). They differ only in **locations** (how far each class centroid is).

Step 4: The Gumbel Race Identity

The Magical Property of Gumbel Distributions

If $G_0, G_1, \ldots, G_{K-1}$ are independent Gumbels with common scale s and locations $\mu_0, \mu_1, \ldots, \mu_{K-1}$:

$$\boxed{P(G_0 < \min_{j \geq 1} G_j) = \frac{1}{1 + \sum_{j \geq 1} \exp\big(-(\mu_j - \mu_0)/s\big)}}$$

This is **exact**. Not approximate. The probability of one Gumbel beating all others is a **logistic function** of the location gaps.

Why this is extraordinary: for most distributions, computing $P(\text{min of } K \text{ things})$ requires intractable K-dimensional integrals. For Gumbels, it factorizes into a simple closed-form sigmoid.

Step 5: Apply to 1-NN

Set $G_0 = M_+$ and $G_j = M_j$:

$$q = P(\text{correct}) = \frac{1}{1 + \sum_{j \neq k} \exp(-\Delta_{k,j}/s)}$$

where $\Delta_{k,j} = \mu_j^{(-)} - \mu_+$ is the "gap" for each competitor.

Step 6: Nearest-class dominance + sparse hierarchy

Two Competition Regimes

Dense (ETF/equidistant): All $K-1$ competitors equidistant $\to \sum = (K-1)e^{-\Delta/s} \to \beta = 1$
Hierarchical (real world): Only $\sqrt{K}$ effective competitors active $\to \beta = 0.5$
Empirical: $\hat{\beta} = 0.478 \approx 0.5$ confirms **sparse competition**

Step 7: Linear gap-to-kappa map

Under isotropic geometry, the gap is approximately linear in κ:

$$\Delta_{\text{nearest}}/s \approx a_1 \kappa_{\text{nearest}} + a_0$$

Combining everything:

The CTI Universal Law — Derived

$$\text{logit}(q_{\text{norm}}) = \underbrace{\frac{a_1}{s}}_{\alpha} \kappa_{\text{nearest}} - \beta \log(K-1) + \underbrace{\frac{a_0}{s}}_{C}$$

Derived: The functional form (logistic in κ, logarithmic in K)
Estimated: The constants (α, β, C)

Neural Collapse: Why the Geometry Is Universal

What is Neural Collapse?

Neural Collapse (NC) is a phenomenon discovered by Papyan, Han, and Donoho (PNAS 2020). At the terminal phase of training:

The Four Properties of Neural Collapse

1. **NC1**: Within-class variability collapses (all points cluster tightly around centroid)
2. **NC2**: Class centroids form an Equiangular Tight Frame (ETF) — maximally equidistant, like a regular simplex
3. **NC3**: Classifier weights align with centroids
4. **NC4**: Predictions converge to nearest-centroid classifier

Why NC matters for CTI

NC provides the geometric **reason** why the law works across architectures:

At exact NC: all class pairs equidistant $\rightarrow \kappa_{\text{nearest}} = \kappa_{\text{spec}}$, within-class isotropic $\rightarrow$ Gaussian assumption satisfied, symmetric Gumbel race $\rightarrow \beta = 1$.

Near NC (real networks): centroids $\approx$ regular simplex, noise $\approx$ shared across classes, bottleneck pair drives classification.

The Key Insight

κ_{nearest} measures **how close** the representation is to Neural Collapse geometry. The law maps this geometric proximity to classification quality.
Different architectures converge to similar NC geometry on the same dataset $\rightarrow$ similar κ $\rightarrow$ similar quality. The architecture determines the *path*; the law governs the *endpoint*.

The Equicorrelation ρ: A Deeper Invariant

What ρ measures

ρ is the average Σ_W-whitened cosine similarity between centroid-difference vectors:

1. Take the within-class covariance Σ_W
2. Whiten: $\Sigma_W^{-1/2}(\boldsymbol{\mu}_j - \boldsymbol{\mu}_k)$ for all pairs
3. Compute pairwise cosine similarities of these whitened vectors
4. Average

- $\rho = 0$: orthogonal competition directions
- $\rho = 1/2$: regular simplex (exact ETF)
- $\rho = 1$: degenerate 1D competition

The remarkable finding: $\rho \approx 0.46$ everywhere

Cross-Modal Equicorrelation — The Tightest Invariant

Modality	ρ
NLP decoders (11 architectures)	0.463
Audio speech (WavLM, HuBERT)	0.455–0.464
Vision (ViT-Base, ResNet50)	0.458–0.467
Mouse visual cortex (Allen)	0.466
NLP encoders (BERT, ELECTRA)	0.446–0.456
Pooled (6 modalities)	**0.462, CV = 1.0%**

Regular simplex prediction: $\rho = 0.500$. Observed: 8% below perfect NC.

α prediction from ρ

From the Gumbel race, the competition effective dimension is $d_{\text{eff,comp}} = 1/(1-\rho)$, and:

$$\alpha_{\text{pred}} = \sqrt{\frac{4}{\pi}} \cdot \frac{1}{\sqrt{1-\rho}} = \frac{1.128}{\sqrt{0.54}} = 1.534$$

Observed: $\alpha = 1.477$. **Error**: +3.8%. **Zero free parameters.**

Caveat

This only works for NLP decoders. For ViT ($\alpha \approx 4.5$) with the same $\rho \approx 0.46$, the prediction gives ∼1.5, not 4.5. A second renormalization factor (related to Σ_W structure) amplifies α for continuous-signal modalities. This is an **open problem**.

The Three Constants: α, β, C

α: The slope (signal amplification)

α converts geometric SNR into classification log-odds. Bigger α = steeper sigmoid.

Signal Type	α	Modalities
Discrete tokens (NLP decoders)	∼1.48	Pythia, Qwen, OLMo, ...
Continuous signals	∼3.3–5.0	ViT (∼4.5), CNN (∼4.0), Audio (∼4.7)
NLP encoders	∼7.7	BERT, DeBERTa, ELECTRA

Within NLP decoders: CV = 2.3% across 12 architectures — effectively a **true constant**. This CV is consistent with LOAO estimation noise (expected ∼2.8%).

β: The competition penalty

β Values

Theory (all $K-1$ competitors active, dense ETF): $\beta = 1.0$
Empirical (sparse hierarchy, $N_{\text{eff}} \approx \sqrt{K}$): $\hat{\beta} = 0.478 \approx 0.5$
Only $\sim\sqrt{K}$ semantically relevant competitors are active. Pre-registered test confirms β=0.5 beats β=1.0 by 10.1% LOAO MAE.

C: The intercept (task difficulty)

C encodes base difficulty of a dataset. It is **NOT universal** — each new dataset requires at least one calibration measurement. LODO (leave-one-dataset-out) mean $r = 0.125$: the intercept can't be predicted across tasks.

Cross-Modal Validation

Vision

ViT-Large on CIFAR-10: $R^2 = 0.964$, same logit-linear form. $\alpha_{\text{ViT}} \approx 4.5$.

ResNet50 on CIFAR-100: $r = 0.749$, $\alpha_{\text{CNN}} \approx 4.4$.

Audio

Seven frozen speech models from 4 architectures on Speech Commands (K=36): $r = 0.898$, $p = 0.006$, $\alpha_{\text{audio}} = 4.669$.

All continuous-signal modalities converge: $\alpha \approx 4$–5.

Biological neural systems

Allen Neuropixels (K=118, 32 mice): 30/32 sessions PASS ($r > 0.50$). Mean $r = 0.736$.

The law form is preserved in biological neurons. The constant differs (A_{bio} is 15–34× smaller), but the geometric skeleton is the same.

Cross-modal ρ: 0.466 in cortex, within 1% of NLP decoders. Near-simplex geometry is **substrate-independent**.

Where biology fails

Human fMRI (NSD, K=12): 1-NN at chance, $r = 0.12$, $p = 0.18$. Raw BOLD voxels lack sufficient SNR. The law requires above-chance 1-NN as a prerequisite.

The Generation Law

The key insight

Next-token prediction (NTP) is V-way classification: the model outputs logits $z_v = \mathbf{w}_v^\top \mathbf{h}(x)$ and the correct token wins the softmax "race." The **same Gumbel race mechanism applies**.

Architecture-Independence Lemma

The generation law depends **only** on the unembedding step $\mathbf{z} = W_U \mathbf{h}(x)$, not on how $\mathbf{h}(x)$ is computed. Therefore: Transformer, SSM, Hybrid → **same law**.

κ from W_U (zero forward passes)

Extract κ directly from the unembedding matrix: normalize rows, compute nearest-neighbor distances among top-1000 tokens. **No inference needed.**

Results

Generation Law Results (22 models)

All models (n=22): $r(\kappa, \log \mathrm{CE}) = -0.55$, $p = 0.008$
Excl. LFM outlier (n=21): $r = -0.70$, $p < 0.001$
Fixed-V (Pythia + Mamba) (n=10): $r = -0.837$, architecture-independent (F-test $p = 0.147$)
Transformers and SSMs lie on the **same regression line**.

Why vocabulary size drops out ($\beta_{\text{gen}} \approx 0$)

At each position, only $K_{\text{eff}} \approx 2$–3 tokens effectively compete (despite V = 32K–152K). The softmax concentrates probability on a few alternatives, so $\log(V-1)$ is irrelevant.

Why R^2 is only 0.49

κ from W_U captures only output geometry, not context quality $\mathbf{h}(x)$. In classification, κ_{nearest} captures the *full* race ($R^2 = 0.955$). The generation law's lower R^2 is **predicted** by the theory.

Causal Evidence

Five tiers establish that this is causal, not just correlation:

Tier 1: Frozen do-interventions

Move one class centroid in embedding space. Nearest-class intervention: $r = 0.899$. Farthest-class (negative control): $r = 0.000$.

Tier 2: Orthogonal factorial

14 focus classes, clean design: Arm A (nearest) $r = 0.899$; Arm B (control) $r = 0.000$; Arm C (farthest) $r = 0.000$.

Tier 3: Cross-modal dose-response

6 unseen architectures (3 text, 3 vision), frozen α, zero refit: 6/6 directional PASS, MAE = 0.010.

Tier 4: Pre-registered confusion matrix prediction

Strongest Causal Test

Fixed $A = 1.054$, $\tau^* = 0.20$, **zero degrees of freedom**:

δ	r	Sign accuracy	p-value
1.0	0.842	92.9%	3.8×10^{-50}
2.0	0.808	**100%**	3.7×10^{-43}
3.0	0.776	**100%**	8.2×10^{-38}

$n = 182$ class-pair predictions per level. **100% sign accuracy** = every predicted direction correct.

Tier 5: Top-m competitor sweep

m=2 competitors recover the LOAO slope: $\alpha_2 = 1.523$ vs $\alpha_{\text{LOAO}} = 1.477$ (2.3% error). Exactly **two near-tie competitors** drive the constant.

Where the Law Fails (Honest Scope)

Hard Failures

1. **Protein LMs** (ESM-2, 7 models): $\alpha = -1.17$, $r = -0.15$. EC classes too heterogeneous.
2. **Human fMRI** (NSD, K=12): 1-NN at chance. Raw BOLD lacks SNR.
3. **Encoder LOAO**: $\text{CV}_\alpha = 0.42$. Encoder α is not family-invariant.

Partial failures: LODO cross-dataset ($r = 0.125$, expected under three-level universality); large-K regime ($r \approx 0.75$ at K=100, below noise floor).

Three-Level Universality Structure

The Hierarchy (Analogy: Equations of State)

Level 1 — Form universality (strongest): The logit-linear structure is EVT-derived and confirmed across NLP, vision, audio, and biological cortex. *Like: $PV = nRT$ holds for all ideal gases.*
Level 2 — Constant within signal type: α is stable within families (NLP decoders CV = 2.3%) but varies across signal types (NLP ∼1.5 vs vision/audio ∼4–5). *Like: van der Waals a, b differ by substance.*
Level 3 — Intercept is task-specific: C encodes dataset difficulty, not predictable across tasks. One calibration measurement required. *Like: ground-state energy differs by system.*

Open Directions

Near-term

1. Expanded generation law validation (25+ models)
2. External replication by independent groups
3. COLM 2026 submission (abstract Mar 26, paper Mar 31)

Medium-term (open theory)

4. **Renormalization theory for α across families**: Why $\alpha_{\text{NLP}} \approx 1.5$ but $\alpha_{\text{ViT}} \approx 4.5$ with same ρ? The answer lies in Σ_W structure.
5. **Non-asymptotic bounds**: Finite-sample Berry–Esseen for Gumbel order statistics.
6. **Multilabel extension**: Current law fails for multilabel (e.g., GoEmotions).

Long-term (paradigm extensions)

7. **Training objective from the law**: Maximize κ_{nearest} during training.
8. **Connection to scaling laws**: If $\kappa \sim C^\gamma$, then $q(C) = \sigma(\alpha C^\gamma + C_0)$.
9. **Universal renormalization**: Derive α from first principles for *any* architecture.

Quick Reference: Key Numbers

The Law

$$\text{logit}(q_{\text{norm}}) = \alpha\,\kappa_{\text{nearest}} - \beta\log(K-1) + C$$

Primary constants (NLP decoders, 12 arch.)

$\alpha = 1.477 \pm 0.034$ (CV = 2.3%), $\beta = 0.478$, $R^2 = 0.955$.

Validation scorecard

Test	Result	Status
12-arch LOAO (NLP)	CV = 0.023	**PASS**
RWKV boundary	$2.887 \in [2.43, 3.29]$	**PASS**
Blind OOD (SmolLM2)	$r = 0.817$, $p = 0.013$	**PASS**
ViT-Large	$R^2 = 0.964$	**PASS**
H8+ holdout (n=77)	$r = 0.879$, MAE = 0.077	**PASS**
Audio (7 models)	$r = 0.898$, $p = 0.006$	**PASS**
Bio (32 mice)	30/32, mean $r = 0.736$	**PASS**
Confusion causal	$r = 0.842$, 100% sign acc	**PASS**
Generation (22 models)	$r = -0.55$ / -0.70	**PASS**
Cross-modal ρ	CV = 1.0%	**PASS**
Protein ESM-2	$r = -0.15$	**FAIL**
Human fMRI	$r = 0.12$, $p = 0.18$	**FAIL**
Encoder LOAO	CV = 0.42	**FAIL**

Zero-parameter predictions

- α from ρ: 1.540 vs 1.477 (+4.3% error)
- Confusion matrix: 100% sign accuracy, zero refit
- LOAO slope from m=2: $\alpha_2 = 1.523$ vs 1.477 (2.3% error)
- K_{eff} from softmax: ∼2–3 ($\beta_{\text{gen}} \approx 0$)

Appendix: The Complete Derivation in One Page

Seven Steps: Geometry → Law

Start: K classes in $\mathbb{R}^d$ with centroids $\{\boldsymbol{\mu}_k\}$ and shared within-class std σ_W.
Step 1 — Define κ:

$$\kappa_{\text{nearest}}(k) = \min_{j \neq k} \frac{\|\boldsymbol{\mu}_j - \boldsymbol{\mu}_k\|}{\sigma_W \sqrt{d}}$$

Step 2 — 1-NN as a race:

$$P(\text{correct} \mid k) = P(M_+ < \min_{j \neq k} M_j)$$

Step 3 — EVT: $M_+, M_j \to \text{Gumbel}(\mu, s)$ with shared scale s.
Step 4 — Gumbel race identity (**exact**):

$$P(M_+ < \min_j M_j) = \left[1 + \textstyle\sum_{j \neq k} \exp(-\Delta_{k,j}/s)\right]^{-1}$$

Step 5 — Nearest-class dominance + sparse hierarchy:

$$\sum \approx (K-1)^{\beta} \exp(-\Delta_{\text{nearest}}/s), \quad \beta \approx 0.5$$

Step 6 — Linear gap-to-κ map: $\Delta_{\text{nearest}}/s \approx a_1 \kappa + a_0$.
Step 7 — Combine:

$$\boxed{\text{logit}(q_{\text{norm}}) = \alpha\, \kappa_{\text{nearest}} - \beta \log(K-1) + C}$$

From Euclidean geometry → EVT → Gumbel race identity → logit law. **QED** (conditional on A1–A3).